Mathematik 1 Komplexe Zahlen

1) a) $z = i$ $\quad z^* = -i$ b) $z = -1$ $\quad z^* = -1$ c) $z = 2+3i$ $\quad z^* = 2-3i$ d) $z = 2e^{i\frac{5}{4}\pi}$ $\quad z^* = 2e^{-i\frac{5}{4}\pi}$

e) $z = 2e^{i\cdot\frac{3}{4}\pi}$

$z^* = 2\cdot e^{i\cdot\frac{3}{4}\cdot\pi}$

2) a)

$z = 3+4i \Rightarrow Re(z) = 3 ; Im(z) = 4 ; |z| = \sqrt{3^2+4^2} = 5 ; \varphi = \tan^{-1}(\frac{4}{3}) \approx 53°$

b) $\frac{z = 1+\sqrt{3}i}{\sqrt{3}+i} = \frac{(1+\sqrt{3}i)\cdot(\sqrt{3}-i)}{(\sqrt{3}+i)\cdot(\sqrt{3}-i)} = \frac{\sqrt{3}-i+3i-\sqrt{3}i^2}{3-\sqrt{3}i+\sqrt{3}i-i^2}$

$= \frac{\sqrt{3}+2i-\sqrt{3}\cdot(-1)}{3-(-1)}$ $\quad |z| = \sqrt{(\frac{1}{2}\sqrt{3})^2+(\frac{1}{2})^2} = \sqrt{\frac{1}{4}\cdot 3+\frac{1}{4}} = 1$

$= \frac{\sqrt{3}+2i+\sqrt{3}}{3+1} = \frac{2\sqrt{3}+2i}{4} = \underbrace{\frac{1}{2}\sqrt{3}}_{=Re(z)} + \underbrace{\frac{1}{2}i}_{=Im(z)}$ i ist nicht Teil des Imaginärteils

$\varphi = \tan^{-1}(\frac{\frac{1}{2}}{\frac{1}{2}\cdot\sqrt{3}}) = \varphi = \tan^{-1}(\frac{1}{\sqrt{3}}) = 60°$

c) $z = 4e^{-i\frac{\pi}{4}}$ $\quad \varphi = -\pi/4 = -45°$ $\quad ; |z| = \sqrt{(4e^{-i\frac{\pi}{4}})\cdot 4e^{+i\frac{\pi}{4}}}$

$z = 4\cdot(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4}))$ $\quad = \sqrt{16} = 4$

$Re(z) = 4\cdot\cos(-\frac{\pi}{4}) ; Im(z) = 4\sin(-\frac{\pi}{4})$ ... was ist $\cos(-\pi/4)$ $\sin(-\pi/4)$??

d) $z = e^{2+i\pi/4}$ $\quad \varphi = \frac{\pi}{4}$ $\quad z = e^2\cdot e^{i\frac{\pi}{4}}$ $\quad ; z = e^2\cdot(\cos(\frac{\pi}{4}) + i\sin(\frac{\pi}{4}))$

$|z|$

$Re(z) = e^2\cdot\cos(\frac{\pi}{4})$ ....

$Im(z) = e^2\cdot\sin(\frac{\pi}{4})$ ....

e) $z = \frac{\sqrt{2}\,e^{i\frac{3}{2}\pi}}{1-i} = \frac{\sqrt{2}\cdot e^{i\frac{3}{2}\pi}}{1\cdot e^{-i\frac{1}{4}\pi}}$

$1-i \Rightarrow \sqrt{1^2+1^2} = \sqrt{2}$ $\quad |z|$

$= \frac{\sqrt{2}\cdot e^{i\frac{3}{2}\pi}}{\sqrt{2}\cdot e^{-i\frac{1}{4}\pi}}$

$= e^{i\frac{7}{4}\pi}\cdot e^{i\frac{1}{4}\pi}$

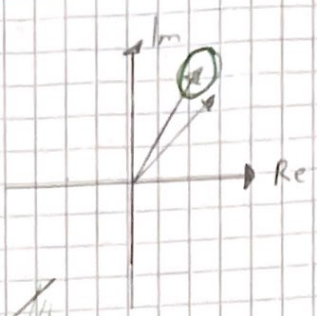
$z = e^{i\frac{7}{4}\pi}$

$z = 1\cdot(\cos(\frac{7}{4}\pi) + i\sin(\frac{7}{4}\pi))$

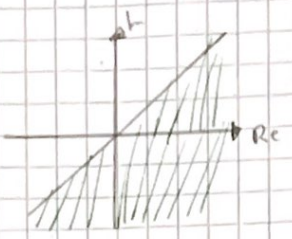
$Re(z) = 1\cdot\cos(\frac{7}{4}\pi) ; Im(z) = 1\cdot\sin(\frac{7}{4}\pi)$

... ...

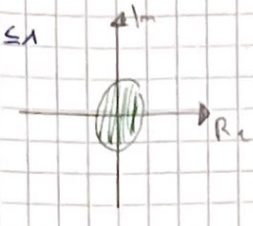
3) a) $|z-(1+i)| = 1$



b) $Re(z) \geq Im(z)$



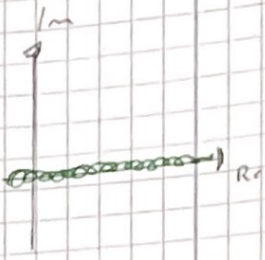
c) $z \cdot \bar{z} \leq 1$



d) $z - \bar{z} = 0$

$z = \bar{z}$

$Im(z) = 0$



4) a) $z = 3+4i$ $\quad |z| = \sqrt{9+16} = \sqrt{25} = 5$; $\varphi = \tan^{-1}(\frac{4}{3}) \approx 53{,}13°$

$Re(z) = 3$; $Im(z) = 4$ $\quad z = 5 \cdot (\cos(53{,}13°) + i \cdot \sin(53{,}13°))$

$z = 5 \cdot e^{i \frac{3}{5}\pi}$

b) $z = -1+i$ $\quad |z| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$; $\varphi = \tan^{-1}(-\frac{1}{1}) + \pi$

$z = \sqrt{2} \cdot (\cos(135°) + i \sin(135°))$ $\quad \varphi = -45° + \pi = 135°$

$z = \sqrt{2} \cdot e^{i \frac{3}{4}\pi}$ $\quad \varphi = -\frac{\pi}{4} + \pi = \frac{3}{4}\pi$

c) $\sqrt{8}\, e^{i\frac{\pi}{4}}$ $\quad \varphi = 45°$ $\quad z = \sqrt{8} \cdot (\cos(\frac{\pi}{4}) + i \cdot \sin(\frac{\pi}{4}))$

$Re(z) = \sqrt{8} \cdot \cos(45°) = a = 2$

$Im(z) = \sqrt{8} \cdot \sin(45°) = b = 2$ $\quad Mz = 2 + 2i$

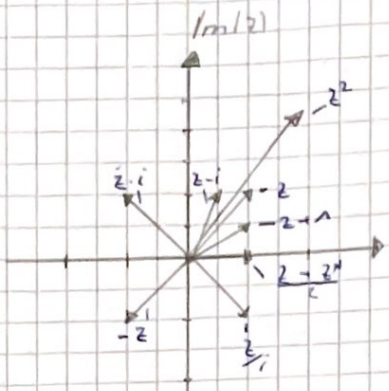
d) $z = 2 \cdot \underbrace{(\cos(\pi) + i \sin(\pi))}_{z = 2 \cdot e^{i \cdot \pi}}$ $\quad Re(z) = 2 \cdot \cos(\pi) = a = -2$

$Im(z) = 2 \cdot \sin(\pi) = b = 0$

$z = -2 + 0i$

5) a) $z-1$ b) $z+1$ c) $z \cdot i$ d) z/i e) $z \cdot e^{i\pi}$ f) z^2 g) $\frac{1}{z} = \frac{1}{|z| \cdot e^{i\varphi}} = \frac{1}{|z|} \cdot e^{-i\varphi}$

h) $\frac{z + z^*}{2} = Re(z)$



6) a) $\cos^2(x) + \sin^2(x) = 1$ $\left[\frac{e^{i\varphi} + e^{-i\varphi}}{2}\right]^2 + \left[\frac{e^{i\varphi} - e^{-i\varphi}}{2i}\right]^2 = 1$

$\frac{e^{i2\varphi} + 2 + e^{-i2\varphi}}{4} + \frac{e^{i2\varphi} - 2 + e^{-i2\varphi}}{4 \cdot i^2} = 1$

$\frac{e^{i2\varphi} + 2 + e^{i2\varphi}}{4} \quad \frac{e^{i2\varphi} - 2 + e^{-i2\varphi}}{4 \cdot (-1)} = 1$

$\frac{2}{4} - \frac{-2}{4} = \frac{4}{4} = 1$ //

b) $\sin(x \pm y) = \sin(x) \cdot \cos(y) \mp \sin(y)\cos(x)$

$\sin(x+y) = \left(\frac{e^{ix} - e^{-ix}}{2}\right) \cdot \left(\frac{e^{iy} + e^{-iy}}{2}\right) + \left(\frac{e^{ix} - e^{-ix}}{2}\right) \cdot \left(\frac{e^{iy} + e^{-iy}}{2}\right)$

$= \frac{e^{i(x+y)} + e^{i(x-y)} - e^{i(-x+y)} - e^{i(-x-y)} + e^{i(x+y)} - e^{i(x-y)} + e^{i(-x+y)} - e^{i(-x-y)}}{4 \cdot 2i}$

$= \frac{2 \cdot (e^{i(x+y)} - e^{i(-x-y)})}{2i} = \sin(x \pm y)$

(7) $\frac{(\cos(x) + i\sin(x))^n}{(e^{i\varphi})^n} = \frac{\cos(nx) + i\sin(nx)}{e^{in\varphi}}$

(8) a) $(\sqrt{2} \cdot e^{i\frac{\pi}{4}})^{802} = \sqrt{2}^{802} \cdot e^{802\frac{\pi}{4}} = 2^{401} \cdot e^{i\frac{\pi}{2}} = i\,2^{401}$

b) $e^{in\pi} = (-1)^n$

c) $\left(\sqrt[5]{3} \cdot e^{i\frac{\pi}{20}}\right)^5 = 3 \cdot e^{i\frac{\pi}{4}}$

d) $e^{i\frac{3}{2}\pi} + 2e^{i\frac{3}{2}\pi} = \cos(\frac{3}{2}\pi) + i\sin(\frac{3}{2}\pi) + 2\cos(\frac{3}{2}\pi) + i\sin(\frac{3}{2}\pi)$

$= -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} + 0 - \frac{2}{2}i = -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}-2}{2}$

(9) $f(1) = 2z^4 - 2z^3 - 2z^2 - 4 \;:\; (z+1) = 2z^3 - 4z^2 - 4z + 2z - 4$

$-(2z^4 + 2z^3)$

$0 - 4z^3 - 2z^2$

$-(-4z^3 - 4z^2)$

$0 \quad 2z^2 - 2z$

$-(2z^2 - 2z)$

$0 \quad -4z - 4$

$-(-4z - 4)$

0

$z = 2$

$2z^3 - 4z^2 + 2z - 4 : (z-2) = 2z^2 + 2$

$-(2z^3 - 4z^2)$

$0 \quad 0 \quad 2z - 4$

$-(2z - 4)$

0

$2z^2 + 2 = 0 \;/-2$

$2z^2 = -2 \;/:2$

$z^2 = -1 \;/\sqrt{}$

$z_{3/4} = \pm i$

(10) a) $z^4 = 81\,e^{i\frac{\pi}{4}} = a$

$z_0 = \sqrt[4]{81} \cdot e^{i\frac{\frac{\pi}{4}}{4}} = 3e^{i\frac{\pi}{16}}$

$z_1 = \sqrt[4]{81} \cdot e^{i\frac{\frac{\pi}{4}+2\pi}{4}} = 3 \cdot e^{i\frac{9}{16}\pi}$

$z_2 = \sqrt[4]{81} \cdot e^{i\frac{\frac{\pi}{4}+4\pi}{4}} = 3 \cdot e^{i\frac{17}{16}\pi}$

$z_3 = \sqrt[4]{81} \cdot e^{i\frac{\frac{\pi}{4}+6\pi}{4}} = 3 \cdot e^{i\frac{25}{16}\pi}$

$z_4 = z_0$

b) $z^3 = \frac{-1+i}{\sqrt{2}} = a = 1 \cdot e^{i\frac{3}{4}\pi}$

$z_0 = 1^3 \cdot e^{i\frac{\frac{3\pi}{4}}{3}} = 1 \cdot e^{i\frac{\pi}{4}}$

$z_1 = 1^3 \cdot e^{i\frac{\frac{3\pi}{4}+2\pi}{3}} = 1 \cdot e^{i\frac{11}{12}\pi}$

$z_2 = 1^3 \cdot e^{i\frac{\frac{3\pi}{4}+4\pi}{3}} = 1 \cdot e^{i\frac{19}{12}\pi}$